ASPECT WAVE LLC

Vessel Measurement Recorder

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Build Guide

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Boards and I2C addresses

Board	Connection	Address	Notes
ESP32-S2 Feather	–	0x36	MAX17048 battery gauge on the board
Adalogger FeatherWing	Feather headers	0x68	Real-time clock (CR1220); microSD chip select pin 10
Ultimate GPS FeatherWing	Feather headers	–	Serial1 (RX / TX), 9600 baud; active antenna on the u.FL
OLED FeatherWing 128 × 64	Feather headers	0x3C	Buttons A = pin 9, B = pin 6, C = pin 5
24LC32 EEPROM	STEMMA QT	0x50	Settings and maxima
BNO085	STEMMA QT	0x4A	Motion sensor
PCF8574 (GPIO)	STEMMA QT	0x20 [0x20–0x27]	Buttons and pot switch
ADS7830 (ADC)	STEMMA QT	0x48 [0x49, 0x4B]	Brightness pot. Not 0x4A: that is the BNO085
BMP388 (optional)	STEMMA QT	0x77 [0x76]	Pressure
SHT30 (optional)	STEMMA QT	0x44 [0x45]	Air temperature and humidity

Addresses in brackets are set with the jumpers on each board; pick the same address in card 17. Each board must have its own address; the defaults all work together.

Feather pins

Pin	Connection
A0 (GPIO 18) A1 (17) A2 (16)	Mark LED Recording LED Fault LED
A3 (GPIO 15)	Free; can be an LED (card 18) or an available analog input
A4 (GPIO 14)	PCF8574 INT (wake-up)
A5 (GPIO 8)	Pot wiper, if wired to the Feather, otherwise an available analog input
5, 6, 9	OLED buttons C, B, A
10	microSD chip select
11, 12	Pot switch and screen button, if wired to the Feather
RX / TX	GPS (through the stack)
SDA / SCL	I2C: STEMMA QT port and the stack

Wiring

Item	Connect	Notes
Pot switch	PCF8574 P0 and GND	Closed = OLED on; open = OLED and LEDs off
Screen button	PCF8574 P1 and GND	Next OLED screen
Mark button	PCF8574 P2 and GND	One mark per press
Cast button	PCF8574 P3 and GND	Hold Cast and press Mark to start or stop a cast
PCF8574 INT	Feather A4	Lets the buttons wake the VMR from Datalogger sleep
Brightness pot (10 kΩ linear)	Wiper to ADS7830 channel 0; ends to VIN and GND	OLED and LED brightness
Mark LED	Feather A0 → 1–2.2 kΩ → LED → GND	
Recording LED	Feather A1 → 1–2.2 kΩ → LED → GND	
Fault LED	Feather A2 → 1–2.2 kΩ → LED → GND	

These are the defaults; card 18 changes them, so write down what you used.

What Header to Solder each Board

Photos

Terminal blocks to I2C breakouts and OLED Featherwing

- 8-pin block on the PCF8574's P0–P7 pads.
- 8-pin block on the ADS7830's A0–A7 pads.

4-pin block on the Qwiic breakout.

Adalogger

- Fit the CR1220 coin cell (+ side up). This keeps the clock running when the VMR is off.
- Insert the microSD card.

[Photo: Adalogger with coin cell and microSD card]

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[Photo: PCF8574 terminal block]

[Photo: ADS7830 terminal block]

Assembly

Enclosure

1. Plan the layout so the OLED is visible through the lid and the BNO085 is mounted flat and square to the box
2. Drill holes for the knob, buttons, power switch, LEDs, USB-C panel mount, both antennas, and PG-7 cable gland

3. Route the GPS antenna cable and Wi-Fi antenna cable and tighten. Connect the external Wi-Fi antenna (w.FL) on the Feather and the active GPS antenna (u.FL) on the GPS FeatherWing
4. Route the SHT30 cable through the PG-7 cable gland and tighten
5. Stack the Adalogger, GPS with CR1220 battery, and OLED FeatherWing on the ESP32-S2 Feather, checking every pin lines up before pushing down. The stack shares power, I2C and the GPS serial line through the headers, so no wires are needed between them
6. Chain the STEMMA QT boards from the OLED's STEMMA QT port, in any order: EEPROM, BNO085, PCF8574 and ADS7830, then the optional BMP388 and SHT30. Arrange them to suit the enclosure. The Feather's STEMMA QT port is switched on by the firmware at start-up; that is normal.
7. Fit and solder the latching power switch between the battery and the Feather
8. Mount the BNO085 flat, chip side up, with its X arrow toward the bow (see Mounting).

[Photo: full wiring diagram]

[Photo: stack assembly]

[Photo: STEMMA QT chain]

Buttons and pot switch

Each button and the pot switch connect between a PCF8574 pin and GND. Use an arcade quick-connect wire for each: push the quick-connect ends onto the switch's two terminals, then screw the other ends into the PCF8574's terminal block, one on the pin (see Wiring table above) and one on GND. No soldering or resistors are needed: the PCF8574 holds the pins high and a pressed button pulls them low.

Wake-up wire: PCF8574 INT to Feather A4. Without it the buttons still work, but they can't wake the VMR from Datalogger sleep (OLED button B always can).

[Photo: PCF8574 button wiring]

[Photo: INT to A4 wire]

Brightness knob

The knob is a potentiometer with an on/off switch on the same shaft. Pot ends to the ADS7830's VIN and GND, the wiper (middle pin) to channel 0. Swap the two ends to reverse which way is brighter.

Without the I2C boards, the controls can go on the Feather itself: screen button to pin 12 and GND, pot switch to pin 11 and GND, and the pot wiper to A5 with its ends on 3V and GND. A5 is used because A0–A4 are on the ESP32's second ADC, which does not read reliably while Wi-Fi runs (as LED outputs they are fine). Never connect the pot to BAT or USB: over 3.6 V on A5 can damage the ESP32-S2.

[Photo: pot wiring]

Indicator LEDs

Each LED has its own Feather pin rather than a PCF8574 pin, so the Fault LED still works when the I2C bus does not. LEDs with resistors toward 2.2 kΩ keep the system darker. A3 is free for another LED or analog input.

Power switch

Splice the latching switch into the battery's red (+) wire; the black (−) wire goes straight to the JST plug. Cut and solder one wire at a time so the red and black never touch.

- Switch off: the battery is fully disconnected and draws nothing.
- With the switch off, plugging in USB still runs the VMR, but the battery only charges with the switch on. Charge mode on the dashboard charges fastest.

[Photo: power switch in battery lead]

Covering board LEDs (optional)

The BNO085, EEPROM, and other I2C boards have green power LEDs that can't be turned off in software. To darken the system, cover them with black electrical tape or a dab of black nail polish.

[Photo: covered LEDs]

Mounting on a Vessel

The VMR measures the box's motion, so the BNO085 must be fixed rigidly to the enclosure, and the enclosure rigidly to the ship. Any wobble shows up as false roll and pitch

- BNO085 flat, chip side up, X arrow toward the bow (Y then points to port)
- Roll shows as S (starboard side down) or P (port side down); in the CSV, + is starboard down. Pitch + is bow up. Rate of turn + is turning to starboard
- Check once: starboard down reads S, bow up reads +, a turn to starboard reads ROT +. If one is backwards, set ROLL_FLIP, PITCH_FLIP or YAW_FLIP to 1 in the sketch or via the Wi-Fi dashboard's Developer Settings
- Enter where the VMR is in card 15: from midships, from the centerline and above the keel.
- With the vessel upright and still, set level zero in card 16 (it uses a 5-s mean of roll and pitch), or calibrate the trim from the draft marks at the dock (card 11)
- Place the GPS antenna with a clear view of the sky
- Location on the ship: on a fixed, rigid surface, e.g. the bridge near the centerline

[Photo: BNO085 mounting]

[Photo: VMR installed on the bridge]

Before you start: Arduino IDE

Needed only to build the firmware from the source code. A VMR that is already running can be updated over Wi-Fi instead (Firmware updates, below)

1. Install Arduino IDE 2.x from arduino.cc
2. Add the ESP32 boards: in File > Preferences (Arduino IDE > Settings on a Mac), paste `https://espressif.github.io/arduino-esp32/package_esp32_index.json` into "Additional boards manager URLs". Then in Tools > Board > Boards Manager, search "esp32" and install "esp32 by Espressif Systems". The VMR is built and tested with version 3.3.12
3. Install the following libraries in Tools > Manage Libraries, accepting the extra libraries when asked

Adafruit SH110X (brings Adafruit GFX and BusIO)	RTClib
Adafruit GPS Library	Adafruit BNO08x
Adafruit MAX1704X	Adafruit BMP3XX
Adafruit SHT31 (also drives the SHT30)	Adafruit PCF8574
Adafruit ADS7830 (with Adafruit Unified Sensor)	SD,

4. Under Tools: Board "Adafruit Feather ESP32-S2", USB CDC On Boot "Enabled", Upload Mode "Internal USB", Partition Scheme "Minimal SPIFFS (1.9MB APP with OTA/128KB SPIFFS)". The default partition scheme (TinyUF2 No OTA) runs but has no room for updates over Wi-Fi

5. Put VMR.ino in a folder called VMR (the folder and file names must match) and open it
6. Plug the Feather in with a USB-C cable that carries data (some are charge-only). Put it into upload mode: double tap RESET. Select the port under Tools > Port and click Upload. The first compile takes a few minutes
7. When the output stops moving and says, “Hash of data verified”, press RESET: the sketch won’t start until you do. The port changes; reselect it and open Tools > Serial Monitor at 115200 baud. The boot list shows OK or FAIL for each part; with only the bare Feather connected, FAIL for the rest is normal

[Photo: Arduino IDE board settings]

If uploads fail

- **“Failed to connect” or no port:** try double-tapping the RESET button so the drive re-connects to the computer, select the new port, upload.
- **“Port busy”:** close the Serial Monitor and anything else using the port, then try again.
- **The port keeps disappearing after RESET:** Unplug from all power for ~10 seconds then try again

Firmware updates

1. In the Arduino IDE: Sketch > Export Compiled Binary.
2. Join the VMR’s Wi-Fi and open the dashboard. In card 20, unlock with the password “vvvv”
3. Choose VMR.ino.bin from the sketch’s build folder (not the .merged, .bootloader or .partitions file) and upload. The VMR restarts, and an open dashboard reloads itself when it is back (if your device is configured to auto-connect to the VMR’s Wi-Fi

A failed update leaves the old firmware running. EEPROM values are kept between restarts and firmware updates

Without the Arduino IDE: a new Feather can be loaded from the single file VMR.ino.merged.bin (same build folder), written at address 0x0 with esptool or Adafruit’s WebSerial ESPTool page in Chrome. Put the Feather in upload mode first (BOOT + RESET).

First start

1. With the knob switch on, depress the power switch,. The OLED shows the Adafruit and Aspect Wave logos, then the start-up checks: each part OK or FAIL. Any FAIL holds the list for 12 s so you can read it.
2. Join the Wi-Fi network “Vessel Measurement Recorder” using password “somestupidcode”
3. Open http://192.168.4.1. The hotspot allows up to 4 connected devices
4. Open Developer Settings. In card 17, press Scan to list every address that answers, tick the parts fitted, check their addresses and Save. The Weather card appears once the BMP388 or SHT30 is on.
5. In card 18, check the pins match the wiring, then Test LEDs.
6. Test each control: the knob changes the OLED brightness and its switch turns the OLED plus LEDs off and on; the screen button changes screens; Mark shows MARK; hold Cast and press Mark for CAST START, and again for CAST STOP.
7. Take the VMR outside or near a window and wait for a GPS fix; the clock sets itself from the GPS.
8. In card 15, enter the ship data and where the VMR is (the keel-to-bridge height and mean draft give the sea-level pressure).
9. Set level zero in card 16, or calibrate at the dock in card 11.

10. Start a short recording in card 7 and check the file in card 9.

How the numbers are worked out

- **Waves:** every 30 s from samples twice a second over the last 17 min (the first after 8.5 min of steady motion; again after a pause or IMU gap; off at the dock). Te is the peak of the pitch spectrum, the pitching period (IMO MSC.1/Circ.1228, 1.6). Tr is the peak of the roll spectrum, the natural roll. The roll period in card 1 times rolls one by one and includes rolling forced at the wave period, so it reads shorter. Te / Tr is flagged within ±15 % of 1 or of ½ (Circ.1228 4.2.3 gives no tolerance)
- **Heave Hs:** 4 × the standard deviation of heave from the vertical acceleration, periods 3–20 s. With the VMR's position in card 15 it is moved to midships, taking off the rise and fall that pitch and roll give a VMR away from midships. A long hull hardly heaves in waves much shorter than itself, so Hs is a guide, not the wave height; long swell comes closest
- **GM:** the IMO rolling-period formula (IS Code 2008, A 2.3): $GM = (2 \cdot C \cdot B / T)^2$, $C = 0.373 + 0.023 \cdot B/d - 0.043 \cdot L/100$, with B the beam, d the mean draft, L the LBP (LOA if none) and T the roll spectrum peak. A cross-check only
- **Ship data checks:** the Cb the displacement implies, $\Delta = Cb \times L \times B \times d \times 1.025$ t/m³ (the quarter-mean draft when a mid draft is given), pointed out when more than 0.06 from the Cb entered; the Cwp the TPI implies, $TPI \times 39.03$ m² / (L × B)
- **Roll statistics:** each full roll counts once; its amplitude is half its peak-to-peak swing about the running mean; the significant roll is the mean of the highest third
- **Trim at the dock:** L × tan(pitch), averaged over the window. Calibrating stores the pitch offset that makes the reading match the drafts; 0.1° is about 1.6 ft on 280 m
- **Weather:** 2-s means kept for 20 min and 30-s means for 3 h. Sea-level pressure adds the sensor's height above the waterline (keel to bridge minus the mean draft). The 3-h change uses a sample every 10 min and counts as steady within ±0.5 hPa. Dew point by the Magnus formula
- **Magnetic heading:** tilt-corrected with the IMU's roll and pitch; the ship's deviation is not removed (still being tested)

Recordings and files

- **Names:** /YYYYMMDD_HHMMSSZ_name.csv, from the UTC time Start was tapped (NOCLK_001… without a clock). A recording cut off by a power loss is not resumed
- **Header:** lines starting with # hold the firmware, ship data, calibration and notes on the columns. Read a file with `pd.read_csv(path, comment='#')`
- **Rows:** GPS columns are filled only on the row after a new fix, and the wave columns only on the row after each new estimate, so nothing is duplicated
- **Card outages:** rows wait in a 256 KB memory buffer (about 3 min at 10 Hz) and are written when the card is back. The first row after is an event "SD RECOVERED after N s (M rows lost)"
- **Location marks:** folder location_mark, one CSV per list, named by the first mark's local time. Local time, ZD and UTC, the position also in decimal degrees, satellites, HDOP, fix age and weather data (if sensors are installed and selected)
- **Downloads:** several files download as one .zip. One day at 10 Hz is about 80 MB; for weeks of data, it's recommended to remove the card from the system

Troubleshooting

Problem	What to do
A part shows FAIL at start-up	Check its STEMMA QT cable is fully seated in at both ends. Scan in card 17 shows which addresses answer. Missing parts are retried every 5 s, so a reseated cable can recover without a restart.
Roll and pitch freeze	The VMR asks the IMU again, then power-cycles the I2C boards by itself; Reset I2C bus in card 19 does it by hand. If the IMU never starts, unplug USB and battery for 10 s
No GPS fix	Check the antenna adapter is seated on the GPS FeatherWing and the antenna can see the sky. The first fix can sometimes take several minutes
GPS antenna shows internal	Only an active antenna shows as external: check the cable is fully-seated. The GPS board's red LED blinks every 1 s while searching, every 15 s with a fix
Mark button doesn't wake it from Datalogger mode	Check the INT-to-A4 wire. OLED button B always wakes it
A phone can't join the Wi-Fi	The hotspot allows 4 connected devices. Restart Wi-Fi or the system in card 19
Dashboard won't load or acts old	Make sure the phone is on the VMR's Wi-Fi (not auto-connecting to another network) then reload the page
Recording LED blinks fast	The microSD card is failing; rows are kept in memory for about 3 min. Check or replace the card
Dashboard slow	Card 19 memory and web timing: slow Updates with short send times mean a weak Wi-Fi signal. Move closer
Can't upload over USB	See If uploads fail, above
Firmware update failed	The old firmware keeps running. Try installing again with the phone closer to the VMR, leaving the Wi-Fi dashboard open. Try the VMR.ino.bin, the app image